

# Percentages — Reverse Percentage & Error

## What Are Percentages?

A percentage is a way to express a number as a part of 100. For example, 50% means 50 out of 100. Think of it like slicing a pizza into 100 pieces!

## ■ Part 1: Reverse Percentage

Reverse percentage helps us find the **original number** before a percentage was added or taken away. Imagine you know the final price — reverse percentage lets you work **backwards** to discover the original price!

### How to Solve Reverse Percentage Problems

|        |                                                                                                                                                                        |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step 1 | For an <b>increase</b> : Add 100% to the given percentage (e.g., $100\% + 20\% = 120\%$ )<br>For a <b>discount</b> : Subtract from 100% (e.g., $100\% - 25\% = 75\%$ ) |
| Step 2 | Convert the percentage to a decimal (e.g., $120\% = 1.2$ , or $75\% = 0.75$ )                                                                                          |
| Step 3 | Divide the final price by the decimal to find the original price                                                                                                       |

### Worked Examples

**Example 1 (Increase):** A shirt costs **Rs 120** after a **20% increase**. What was the original price?

Step 1:  $100\% + 20\% = 120\%$

Step 2:  $120\% = 1.2$

Step 3:  $\text{Rs } 120 \div 1.2 = \text{Rs } 100$

✓ The original price was Rs 100!

**Example 2 (Discount):** A bicycle costs **Rs 1,800** after a **10% discount**. What was the original price?

Step 1:  $100\% - 10\% = 90\%$

Step 2:  $90\% = 0.90$

Step 3:  $\text{Rs } 1,800 \div 0.90 = \text{Rs } 2,000$

✓ The original price was Rs 2,000!

## ■ Part 2: Percentage Error

Percentage error tells us **how far off** our estimate is from the actual value. It helps us measure how accurate our guess or measurement was. The smaller the error, the better our estimate!

### Percentage Error Formula

$$\% \text{ Error} = ( |\text{Estimated Value} - \text{Actual Value}| \div \text{Actual Value} ) \times 100\%$$

#### Key Points:

- Always use absolute value (ignore negative signs)
- Always divide by the **actual** value
- Multiply by 100 to get the percentage

### Worked Example

**Example:** Riya estimated **45 mangoes** in a box. Actual count was **50**. Find the percentage error.

$$\text{Error} = |45 - 50| \div 50 \times 100\%$$

$$= 5 \div 50 \times 100\%$$

$$= 10\%$$

✓ Riya's estimate was 10% off!

## ⇒ ■ Practice Problems

| # | Type      | Problem                                                                   | Answer                                                                               |
|---|-----------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1 | Reverse % | A watch costs Rs 450 after a 10% increase. Find the original price.       | $100\% + 10\% = 110\% = 1.1 \rightarrow \text{Rs } 450 \div 1.1 = \text{Rs } 409.09$ |
| 2 | % Error   | Priya estimated 60 students in a hall. Actual count was 55. Find % error. | $ 60 - 55  \div 55 \times 100\% = 5 \div 55 \times 100\% \approx 9.09\%$             |

|   |                  |                                                                       |                                                                                          |
|---|------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 3 | <b>Reverse %</b> | A bag costs Rs 680 after a 15% discount. What was the original price? | $100\% - 15\% = 85\% = 0.85 \rightarrow \text{Rs } 680 \div 0.85 \approx \text{Rs } 800$ |
| 4 | <b>% Error</b>   | Arjun guessed the length of a rope as 8 metres. Actual = 7.5 m.       | $ 8 - 7.5  \div 7.5 \times 100\% = 0.5 \div 7.5 \times 100\% \approx 6.67\%$             |
| 5 | <b>Reverse %</b> | A toy costs Rs 180 after a 20% discount. What was the original price? | $100\% - 20\% = 80\% = 0.80 \rightarrow \text{Rs } 180 \div 0.80 = \text{Rs } 225$       |
| 6 | <b>% Error</b>   | Estimated weight: 52 kg. Actual: 50 kg. Find percentage error.        | $ 52 - 50  \div 50 \times 100\% = 2 \div 50 \times 100\% = 4\%$                          |

## ■ Key Takeaways

|   |                                                                         |
|---|-------------------------------------------------------------------------|
| ■ | Reverse percentage finds the original value before a % change           |
| ■ | Always work with 100% as your base                                      |
| ■ | Percentage error measures accuracy of estimates                         |
| ■ | Formula: $ \text{Est} - \text{Actual}  \div \text{Actual} \times 100\%$ |
| ■ | Practice makes perfect!                                                 |

**Keep practising percentages! You're doing amazing.**  
Remember: Maths is fun when you try!